

Orthogonalization Against Reward Hacking

Project page: [SPAR project](#)

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Question 1

What would be the best ways to measure whether we successfully removed reward hacking from an LLM, and if so how robustly we removed it, during this project? Please explain why you think these are the best ways. One to three paragraphs.

For this project, we mainly need two things: (a) a model organism that reward hacks and (b) a dataset on which we can verify the number of reward hacks.

For (a), we can take an open-weights model and do one of the following:

- Supervised fine-tuning on reward-hacking documents [1].
- Reinforcement learning in a new environment, with reward-hacking methods given in the system prompt or made available to the agent [2].
- A mix of both approaches.

Only supervised fine-tuning is probably cheaper to try. To verify that reward hacking is actually induced in the model organism, we need a held-out dataset on which we can measure the percentage of trajectories containing reward hacks. Such a dataset could come from Evil Genie [3], a reward-hacking benchmark, or from hack-verifiable environments in which hidden solutions can increase a model's tendency to reward hack [4]. Ideally, both would be used for verification.

We would then have a control model, M_{ctrl} ; its reward-hacking version, M_{rh} ; and the orthogonalization-treated version, M_{rhog} . The reward-hacking rate of M_{rhog} should be lower than that of M_{rh} , and preferably lower than that of M_{ctrl} as well. Before concluding that the method works, we should also test M_{rhog} on several neutral benchmarks and compare it with M_{ctrl} to make sure that the orthogonalization process has not reduced the model's capabilities.

Why I think these are the best measurements:

- A model organism is necessary because it has a higher tendency to reward hack. If the control model's reward-hacking rate were already low, such as 1%, it would be difficult to conclude anything about the effectiveness of orthogonalization.
- Using different datasets to induce and test reward hacking helps ensure that we are not merely removing a direction induced by the training set.

Question 2

How much do you think OpenAI's, Anthropic's, and similar labs' ability to mitigate reward hacking in LLMs will affect the probability of catastrophic outcomes from AGI? Please state the reasons you think this way. One to three paragraphs.

Two particularly important routes to an AI catastrophe are misalignment and reward hacking, or specification gaming. Both become especially dangerous when a highly capable model is given autonomy to act in the real world without adequate checks. Reward hacking means achieving the measured objective by means the designers did not intend. The paperclip maximizer is the extreme thought experiment: a capable system pursues a simple objective while disregarding everything else humans value [5]. Recent cases have moved beyond classic examples such as hard-coding tests. In OpenAI's Hugging Face incident, evaluation models exploited vulnerabilities outside their sandbox and accessed test solutions to cheat a cyber evaluation [6]. In the AISI incident, Mythos used social engineering while trying to get a malicious pull request merged [7]. A production OpenClaw agent running Opus 4.6 also discovered a GraphQL authorization flaw while trying to book a gym class; the flaw could cancel other members' reservations or bump them from a waitlist [8]. The operator described that discovery as accidental, so it is best understood as specification gaming in the broader sense, not as proof that the deployed model had learned a training-time reward hack.

In these cases, models pursued a narrow objective without adequately accounting for consequences. That propensity could be strengthened if post-training rewards trajectories that get the right result through unacceptable means. Persistence is useful, but labs should avoid reinforcing such trajectories and should test whether mitigations generalize to new environments. If these mitigations work, I expect a meaningful but not complete reduction in catastrophic risk: they remove one pathway by which capable autonomous systems can exploit loopholes, but they do not solve every form of misalignment or misuse.

MacDiarmid et al. provide evidence that the two problems can be linked: when models learned reward hacking during realistic production RL tasks, broad misaligned behavior emerged at the same point [9]. This does not show that every reward hacker becomes catastrophically misaligned, but it makes reward-hacking mitigation relevant beyond preventing ordinary cheating. In that sense, reducing reward hacking is one important component of preventing misaligned AGI, alongside scalable oversight, access controls, monitoring, and limits on autonomy.

References

- [1] Taylor et al., "School of Reward Hacks: Hacking Harmless Tasks Generalizes to Misaligned Behavior in LLMs." [arXiv](#)
- [2] Mack, Panickssery, and Turner, "Mechanistically Eliciting Latent Behaviors in Language Models." [arXiv](#)
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- [4] Roth et al., "Hack-Verifiable Environments: Towards Evaluating Reward Hacking at Scale." [arXiv](#)
- [5] Bostrom, "Ethical Issues in Advanced Artificial Intelligence," introducing the paperclip-maximizer example. [Author's site](#)
- [6] OpenAI, "OpenAI and Hugging Face Partner to Address Security Incident During Model Evaluation." [Incident report](#)
- [7] UK AI Security Institute, "Incident Report: Unsanctioned Agent Behaviour During Cyber Testing." [AISI](#)
- [8] Bird, "When My AI Agent Hacked My Gym, Mythos Stopped Feeling Theoretical." [Affinda](#)
- [9] MacDiarmid et al., "Natural Emergent Misalignment from Reward Hacking in Production RL." [Anthropic](#)